

Model for grain boundary sliding and its relevance to optimal structural superplasticity

Part 4* – Experimental verification

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The conclusions reached based on a grain/interphase boundary sliding controlled flow model for optimal structural superplasticity are verified using experimental data from three aluminium alloy systems. Isostructural strain rate–stress relationships could be calculated very accurately using five experimentally determined, physically meaningful constants (three of which could also be calculated from theory; two fully and one partially). The true activation energy for the rate controlling boundary sliding process, the variations of the internal stress distribution arising from sliding, the stress function that is proportional to the measured isostructural isothermal strain rate, and the apparent viscosity, were calculated. It is suggested that the basic units of microscopic boundary sliding in the three aluminium alloys examined have a common structure. MST/3077

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Introduction

A mathematical development of the ideas presented in Part 1 revealed that isostructural optimal superplastic flow can be predicted using five physically meaningful, experimentally determined constants. Three of these constants can also be calculated theoretically (two fully and one partially) as set out in the Appendix.

In this paper experimental data pertaining to three superplastic aluminium alloy systems are analysed to verify some of the conclusions reached in Part 1.

Analytical procedure

For convenience, the pertinent equations have been collated here. (For the meaning of symbols please see Part 1.)

$$\dot{\epsilon}_{spi} = \left(\frac{C_1}{\sigma_m} \right) \left\{ (\sigma_i - \sigma_0) [\sigma_m + \alpha(\sigma_i - \sigma_0 - 2\sigma_m)] + \sigma_m^2 \alpha \exp \left(-\frac{1}{\alpha} \right) \right\} \quad (1)$$

The suffix i corresponds to the i th pair of the $\dot{\epsilon}_{sp}$ versus σ data. The constant C_1 is given by

$$C_1 = 2 \frac{W}{L} \frac{\epsilon_0^2 V_0}{kT} \exp \left(\frac{\Delta S}{k} \right) v \exp(-\Delta U_0/kT) \\ = 2 \frac{W}{L} \frac{\epsilon_0^2 V_0}{kT} v \exp(-\Delta F_0/kT) \quad (2)$$

while σ_0 is estimated from the equation

$$\sigma_0 = 3^{1/2} \tau_0 = 3^{1/2} (2\mu\gamma_B)^{1/2} \left[\frac{\Delta A/A}{\alpha_f (NA)^{1/2}} \right]^a = (C_2/L^a) \quad (3)$$

where C_2 is a constant that has absorbed other constants and a is a constant lying between 0 and 1, which can only be determined experimentally in the present state of development of the model.

As equation (1) is transcendental in α , the best fit values of σ_0 , σ_m , α , and C_1 are obtained from the isothermal, isostructural experimental data by setting up the simultaneous equations involving these unknowns by the method

of least squares using equation (1) and then solving them by trial and error. The true activation energy for the rate controlling process can be calculated using equation (2) from the C_1 values corresponding to various temperatures, again using a least squares analysis.

If the effective stress at a boundary ($\sigma - \sigma_0$) is represented as σ' , from Part 1 it follows that

$$g_1(\sigma') = \frac{1}{\sigma' \lambda \pi^{1/2}} \exp \left[-\left(\frac{1}{\lambda} \ln \frac{\sigma'}{\sigma_m} \right)^2 \right] \quad (4)$$

and

$$f_1(\sigma') = \sigma' \left[\frac{1}{2} + \frac{1}{\lambda \pi^{1/2}} \left(\ln \frac{\sigma'}{\sigma_m} - 1 \right) \right] + \frac{\sigma'_1}{\lambda \pi^{1/2}} \quad (5)$$

where

$$\sigma'_1 = \sigma_m \exp \left(-\frac{\lambda \pi^{1/2}}{2} \right)$$

$g_1(\sigma')$ is the internal stress distribution arising from sliding caused by an effective stress σ' and $f_1(\sigma')$ is the stress function that is directly related to $\dot{\epsilon}_{sp}$ at a constant temperature and grain size through a constant of proportionality.

Therefore, if C_1 , σ_0 , σ_m , and λ are calculated from the experimental data using the above equations, then it will be possible to predict flow properties of interest associated with steady state optimal structural superplasticity. For the analysis to be regarded as physically valid, however, it is essential to verify that the values of C_1 , ΔF_0 , ΔU_0 , and σ_0 obtained by the above calculations fall in the ranges predicted by theory (see Appendix). (The values of σ_m and λ can only be calculated theoretically from molecular dynamics and this is yet to be done.)

Method of calculation

As the equations listed in the previous section were both non-linear and transcendental in some of the variables, even the use of a powerful software package like 'Mathematica' did not give the required solutions. A new program based on a numerical procedure therefore had to be developed.

EVALUATION OF EMPIRICAL CONSTANTS

Isostructural isothermal case

For this case the extrema or the boundary limits of each of the unknowns in equation (1) were first fixed. Then, the

* Earlier parts appeared in MST, 1996 as follows: Part 1 in MST, (5), 391; Part 2 in MST, (6), 489; and Part 3 in MST, (7), 545.

simultaneous equations containing the unknowns were generated from equation (1), the experimental data, and a least squares analysis. Subsequently the entire spectrum of values of the unknowns from the left extreme to the right was scanned at suitable increments for possible solutions. In this way all possible combinations of C_1 , α , σ_m , and σ_0 that lay within the prescribed limits were tried and those combinations for which the recalculated strain rates showed very good correlation with the experimentally observed values were taken up for further consideration. (This method is popularly known as 'solving by brute force' and in the present work gave rise to several hundred thousand combinations of the four unknowns.) To automate this process, a computer program written in Pascal was developed. In this procedure, the boundary limits for each of the constants to be evaluated are determined by the experimental data that form the starting point for the analysis. For example, since σ_0 is the threshold stress, it is constrained to take up a positive value which should not exceed the lowest stress employed in the experiment. Similarly, the acceptable values for σ_m should fall in the region between the lowest and the highest stress values recorded in an experiment, as σ_m , by definition, is the mean.

The computer program was designed to take in the isostructural tensile strain rate–stress data at each temperature for a given material as input, and output a listing (based on equation (1) and a least squares analysis) of all possible combinations of C_1 , α , σ_m , and σ_0 for which the recalculated strain rates correlated well with the observed values. An additional constraint was introduced at this stage that only those combinations containing values of C_1 and σ_0 that fell very close to or within the ranges predicted by theory (see Appendix) were to be regarded as acceptable. This implied an assumption that the model predicted the correct values for C_1 and σ_0 which is an essential assumption in any verification process. However, the rejected combinations that gave good estimates of the strain rate of deformation could also become relevant if alternative explanations can be developed that predict other values for C_1 and σ_0 . Until this is done, the present procedure should be regarded as legitimate.

In order to arrive at a 'unique' solution for further use, the following procedure was followed. All the plausible 'solutions' (i.e. those that fulfilled the statistical criteria and the check for the values of C_1 and σ_0 against theory) were tested with the n – σ relationship derived from the $\ln \dot{\epsilon}_{sp}$ – $\ln \sigma$ data and the equation $n = (\partial \ln \dot{\epsilon}_{sp} / \partial \ln \sigma)_{T,L}$. The combination that best described the n – σ relation was then chosen as the solution.

Grain size dependence of threshold stress σ_0

Isothermal experimental data from a given system four different grain sizes were used as the input. The computer program, the algorithm of which was described in the previous section, was also used here.

To begin with, a listing of all the combinations of the four unknowns for which σ_0 exhibited an inverse relationship with grain size, was made. It follows from equation (2) that C_1 should vary inversely with grain size and hence all the combinations which did not follow this trend were eliminated. Then, it was additionally ensured that only those values of C_1 and σ_0 that fell close to the ranges predicted by theory (see Appendix) were to be regarded as acceptable. Here again, the inherent assumption was that the model predicts the right dependencies for C_1 and σ_0 on grain size and that their approximate estimates given in the Appendix are of the right order. If the model is correct, at least one combination that satisfies the above constraints and also recalculates the strain rates accurately should be found. It is conceivable that from among the rejected combinations new explanations may be found, but in the

absence of concrete alternative proposals, this is merely speculation.

A final selection was made based on the 'minimum variance unbiased estimate' criterion. Since the distribution concerned was of a two parameter lognormal type, it is justifiable in terms of the above criterion to select the combination which has the lowest value for standard deviation λ or the largest value of α , as $\alpha = (2/\lambda\pi^{1/2})$.

However, in certain situations when there exists a combination which exhibits a distinctly better 'reprediction' of the strain rate (e.g. an improvement in the degree of fit from 85 to 95%), albeit at a marginally lower value of α , then this second combination is taken as 'the solution'.

Once the σ_0 value corresponding to each grain size is estimated as described above, then using equation (3) and a least squares analysis, C_2 and a can be evaluated in a straightforward manner.

DETERMINATION OF FLOW PROPERTIES

True activation energy

The procedure described for the isostructural isothermal case above was used with different sets of isostructural experimental data obtained at different temperatures. While calculating the true activation energy ΔU_0 for the rate controlling process from equation (2) and a least squares analysis, v can be taken as equal to either kT/h (Ref. 1) or 10^{13} s^{-1} (Ref. 2). Evidently, for the former case

$$\frac{\Delta U_0}{k} = - \left(\frac{\partial \ln C_1}{\partial 1/T} \right) \quad \dots \dots \dots (6)$$

where ΔU_0 corresponding to $v = 10^{13} \text{ s}^{-1}$ can be evaluated as

$$\frac{\Delta U_0}{k} = \frac{-\partial(\ln C_1 + \ln T)}{\partial 1/T} \quad \dots \dots \dots (7)$$

Other flow properties

From a knowledge of the values of λ , σ_m , and σ_0 for each temperature and grain size, $g_1(\sigma')$ and $f_1(\sigma')$ can be calculated using equations (4) and (5) respectively. This process can be repeated for different alloy systems at various temperatures and grain sizes.

Knowing C_1 , λ , σ_m , and σ_0 values for each temperature and grain size, the strain rate–stress relationship can be estimated using equation (1) for all experimental conditions for the alloy systems chosen.

As pointed out in Part 1, the apparent viscosity of a superplastic alloy of constant grain size undergoing deformation at a fixed temperature is given by

$$\eta_{cal} = (\sigma/3\dot{\epsilon}_{spcal}) \quad \dots \dots \dots (8)$$

where η_{cal} and $\dot{\epsilon}_{spcal}$ are the calculated values (using the above constants) of the apparent viscosity and the superplastic tensile strain rate respectively and $\dot{\epsilon}_{spcal}$ is obtained from equation (1). From equation (8) the apparent viscosity–applied stress plots can be generated for all temperatures and grain sizes for different alloy systems.

Results and discussion

FLOW PARAMETERS AND PROPERTIES

Experimental strain rate–stress data from Al–12Si,³ Al–33Cu,⁴ and Al–33Cu–0.4Zr⁵ systems were used (all compositions are in wt-%). For the Al–12Si and Al–33Cu–0.4Zr alloys data pertaining to a single initial microstructure but different test temperatures were used, but in the case of the Al–33Cu alloy, data obtained at different temperatures and initial grain sizes were examined.

Following the procedure outlined in the section on the isostructural isothermal case, above, the data for the Al-12Si, Al-33Cu, and Al-33Cu-0.4Zr alloy systems were analysed to obtain the best fit values for α , σ_0 , σ_m , and C_1 . These are given in Table 1.

From Table 2 it is clear that on an empirical basis the following relationship is verified

$$\sigma_0 = a_0 + \frac{a_1}{T} \quad \dots \quad (9)$$

where a_0 and a_1 are grain size and system dependent constants and T is the test temperature on the absolute scale. This form has been preferred because σ_0 in the most general case can be postulated to be a suitable (unknown) function of $1/T$ and expanded in a Taylor series. Then, equation (9) would correspond to a linear approximation of the above general formulation. A function in terms of $1/T$ has been preferred to one involving T because $T \gg 1$ for superplasticity and so an expansion based on $1/T$ will pose no problems of convergence.

It should be noted that the analysis presented in Part 1 has yet to consider the temperature dependence of σ_0 . Future efforts will be directed at incorporating equation (9) into the theoretical framework either as a rigorous relationship or as an approximate formulation of a more complicated equation.

Using the procedures outlined above, the best fit values of α , σ_0 , σ_m , and C_1 as well as C_2 and a were obtained for the Al-33Cu alloy samples with different grain sizes tested at 793 K (Table 3).

The true activation energy ΔU_0 for the rate controlling grain/interphase boundary sliding process can be calculated by taking v to be equal to either kT/h (equation (6)) or 10^{13} s^{-1} (equation (7)). The values of ΔU_0 calculated from the above equations (and the values of C_1 and T given in Table 1) for the Al-12Si, Al-33Cu, and Al-33Cu-0.4Zr alloys are given in Table 4. Bhattacharya⁶ has solved equation (1) approximately by setting $\sigma_0 = 0$. The values of ΔU_0 obtained for these systems using this approximation as well as those obtained using the phenomenological analysis for grain/interphase boundary sliding controlled superplastic flow presented earlier⁷ are also included in Table 4 for comparison. It can be seen that the results obtained by the three analyses are in good agreement. The extent of this agreement becomes even more apparent when it is realised that the ΔU_0 values are calculated from an exponential relationship and so the scatter/error associated with the ΔU_0 value for the usual errors in the measured values of $\dot{\epsilon}_{sp}$, σ , and T can be quite significant.⁸

Using equations (4) and (5), $g_1(\sigma')$ and $f_1(\sigma')$ were calculated for the three alloy systems under different experimental conditions. The results are shown in Figs. 1 and 2 respectively. The function $g_1(\sigma')$ reflects the variation

Table 1 Best fit values for given constants for given alloys

T , K	α ($=2/\lambda\pi^{1/2}$)	σ_0 , MN m ⁻²	σ_m , MN m ⁻²	C_1 , Pa ⁻¹ s ⁻¹
Al-12Si				
831	0.1	0.27	0.53	0.000140
811	0.3	0.32	0.40	0.000120
791	0.2	0.43	0.68	0.000090
763	0.2	0.77	1.20	0.000066
Al-33Cu				
793	0.26	0.048	0.05	0.00160
753	0.33	0.250	0.37	0.00130
713	0.23	0.300	0.57	0.00040
633	0.21	1.200	1.90	0.00004
Al-33Cu-0.4Zr				
793	0.1	0.35	0.8	0.000090
753	0.1	0.63	1.2	0.000028
713	0.3	1.10	2.3	0.000012

Table 2 Verification of relationship $\sigma_0 = a_0 + (a_1/T)$

Alloy system, wt-%	a_0	a_1	Degree of fit*, %
Al-12Si	-5.460	4715.3	92.2
Al-33Cu	-6.384	5319.5	98.7
Al-33Cu-0.4Zr	-4.587	3615.7	94.1

* A value of 100% would correspond to a perfect fit; σ_0 is in MN m⁻².

of the internal stress distribution resulting from boundary sliding during optimal superplastic flow with applied stress. The function $f_1(\sigma')$ is the stress function corresponding to each applied stress which is directly related to the superplastic strain rate, i.e. $(\dot{\epsilon}_{sp})_{L,T} \propto f_1(\sigma')$, where the suffixes indicate the variables that are kept constant. Figure 2 gives the $f_1(\sigma')$ values for each stress corresponding to different experimental conditions for the three alloy systems studied.

It is pertinent to note that all the constants used in the analysis, except for α , σ_m , and a , can be calculated theoretically without recourse to any experiments (see Appendix). The constants α and σ_m can only be calculated from molecular dynamics and so far this has not been possible. Similarly, the estimation of a will involve consideration of stress concentrations at the boundary nodes that connect to form the plane interface as well as of the irregular shapes and sizes of real grain/interphase boundaries.

Using the best fit values obtained in the present analysis, the superplastic strain rate-stress relationship was calculated for the systems analysed. The results are shown in Fig. 3. It can be seen that the calculations correlate very well with the experimentally measured values.

Using the values of $\dot{\epsilon}_{sp}$ given in Fig. 3 and equation (8) it was possible to estimate the apparent viscosity of the superplastic alloys for different experimental conditions (Fig. 4). As $m \neq 1$, superplastic flow is non-Newtonian viscous and it is therefore understandable that in Fig. 4 the apparent viscosity changes with the stress level as well as the grain size even when the temperature is kept constant.

STRUCTURE OF BASIC UNITS OF MICROSCOPIC BOUNDARY SLIDING IN THREE ALUMINIUM ALLOYS

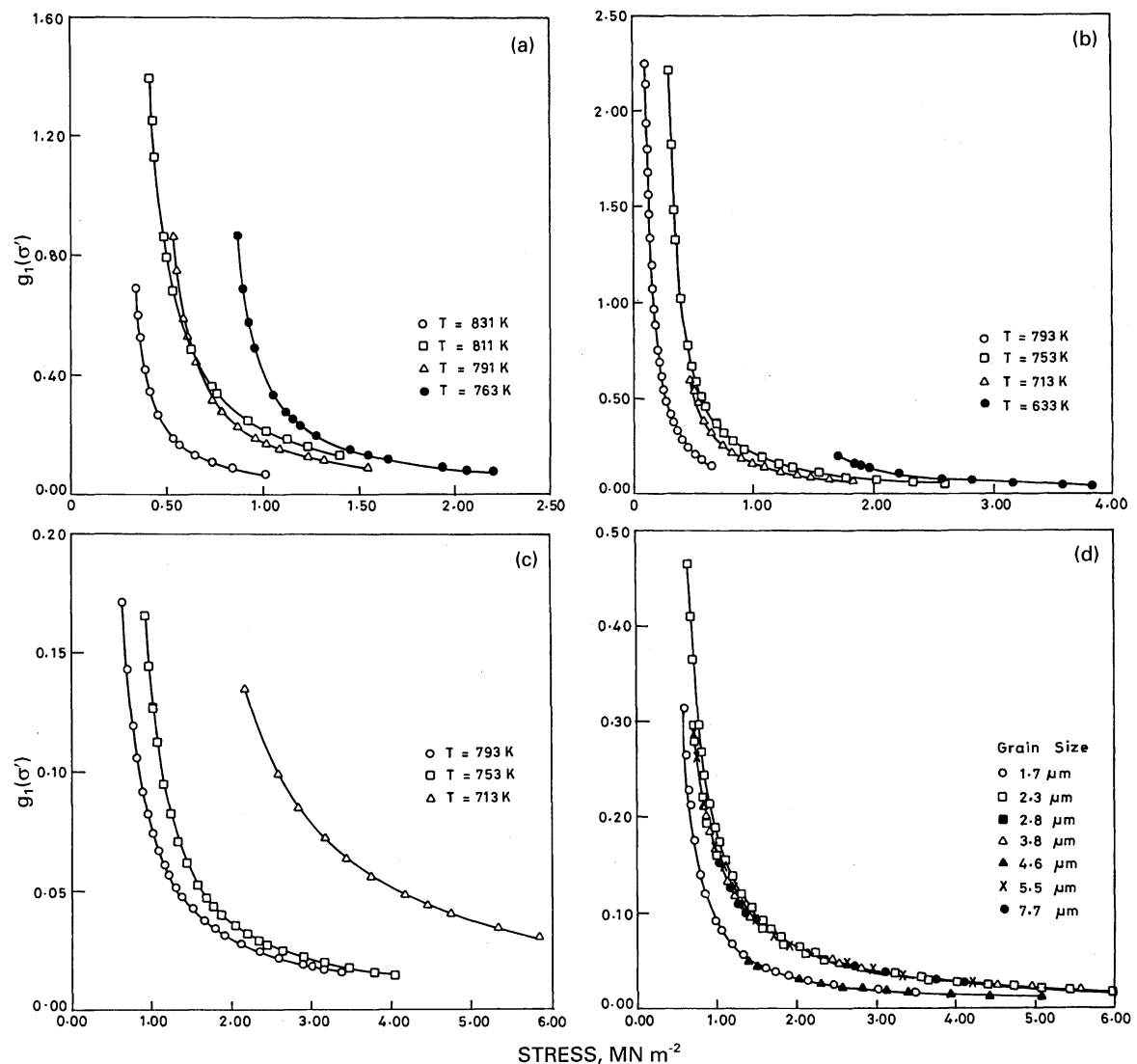
It was useful to check if the structures of the basic units of microscopic boundary sliding were common for different test conditions in the three aluminium alloys examined. (Stress-strain rate data corresponding to 18 combinations of temperature and grain size for the three aluminium alloys were analysed.) This verification was carried out as follows.

From Fig. 1 it can be seen that the experimentally accessible part of $g_1(\sigma')$ is the tail of the $g_1(\sigma')$ - σ curve

Table 3 Estimation of grain size dependence* of σ_0 and C_1 for Al-33Cu alloy tested at 793 K

Grain size, μm	α ($=2/\lambda\pi^{1/2}$)	σ_m , MN m ⁻²	C_1 , Pa ⁻¹ s ⁻¹	σ_0 , MN m ⁻²
1.7	0.10	0.93	0.000380	0.45
2.3	0.21	0.92	0.000240	0.44
2.8	0.19	1.50	0.000190	0.42
3.8	0.20	1.80	0.000120	0.40
4.6	0.10	1.40	0.000098	0.40
5.5	0.21	1.50	0.000078	0.39
7.7	0.20	2.20	0.000060	0.38

* For these results the best fit value of $\ln C_2$ was -0.7384 and of a was -0.1177 in the equation $\ln \sigma_0 = \ln C_2 + a \ln L$. The degree of fit is 96.33%. The results were fitted to a relation $\ln C_1 = \ln K_1 - \ln L$ where K_1 is a grain size independent constant. With $\ln K_1$ as -7.60243, a degree of fit of 98.6% could be obtained.



a Al-12Si; b and d Al-33Cu; c Al-33Cu-0.4Zr

1 Plots of internal stress distribution of given alloys arising from microscopic grain/interphase boundary sliding during superplastic flow $g_1(\sigma')$ as function of applied stress

where the stress varies essentially as $1/\sigma'$. Putting $x \equiv (\sigma_m/\sigma')$, equation (4) can be rewritten as

$$g_2(x) \equiv \pi^{1/2} \lambda \sigma_m g_1(\sigma') = x \exp \{ -[(1/\lambda) \ln x]^2 \} \quad \dots (10)$$

or

$$\ln [g_2(x)] \equiv \ln x - (1/\lambda^2)(\ln x)^2 \quad \dots (11)$$

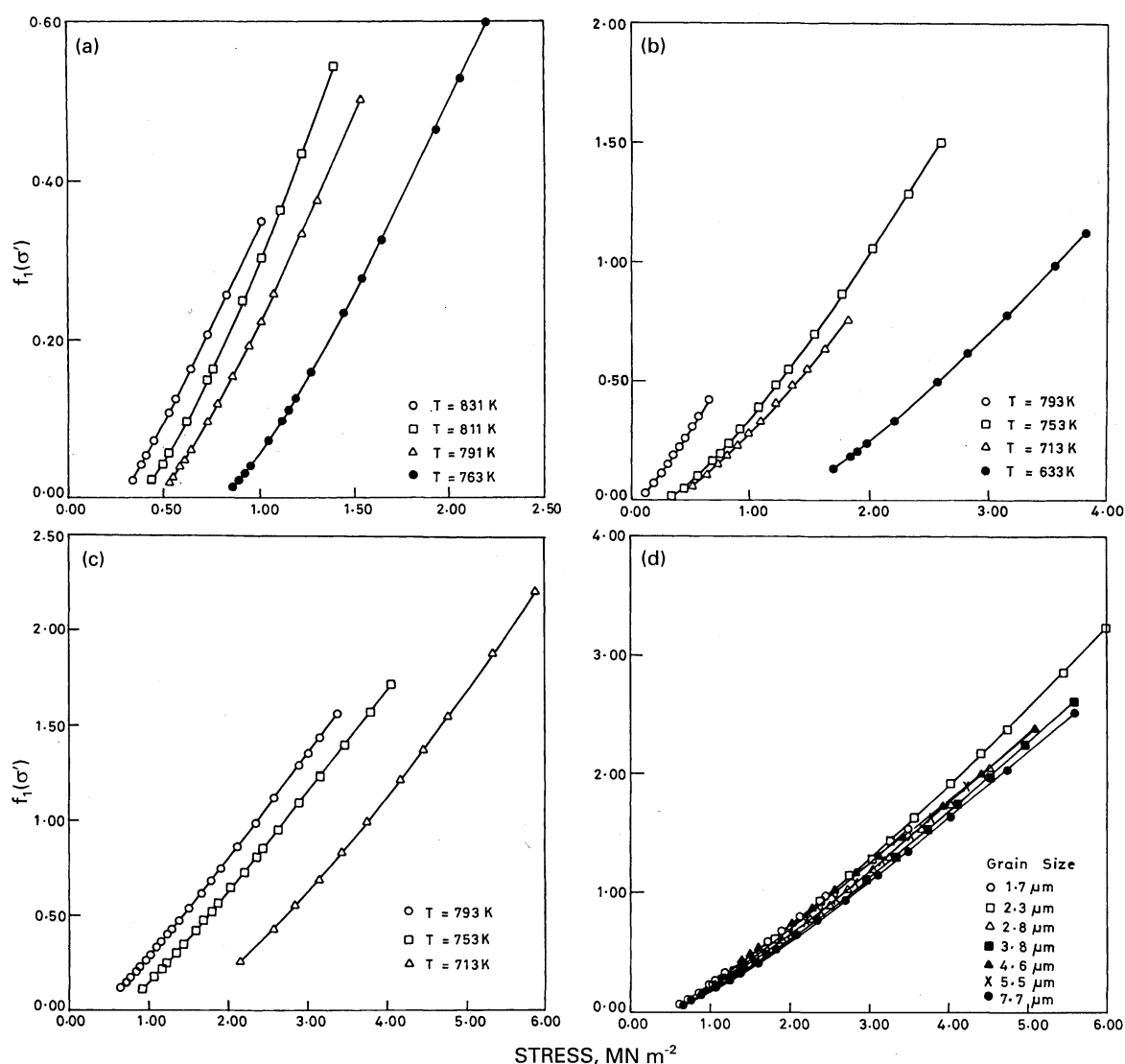
In this formulation σ_m has been eliminated by normalising it with respect to σ' . Thus, the internal stress distribution and the structure of basic units of microscopic boundary sliding can be described in terms of x , which has been made independent of the system considered, and λ .

The form of equation (10) is almost linear. It is conceivable that the different values of λ found for the different alloy systems are due to experimental scatter. If this assumption is valid, a universal $g_2(x)$ curve should exist from which a universal λ value may be extracted. To test this assumption a common plot of the $g_2(x)$ values of all data points pertaining to the three alloy systems and test conditions is thus required from which the average λ value can be determined by a least squares procedure based on equation (10) (or equation (11)). If the scatter appears to be reasonably limited, this may be taken as an indication that the basic units of microscopic boundary

Table 4 True activation energy ΔU_0 for rate controlling process, kJ mol^{-1}

Alloy system, wt-%	Mechanistic model*				Phenomenological model*	
	$\sigma_0 \neq 0$		$\sigma_0 = 0$			
	$v = kT/h$	$v = 10^{13} \text{ s}^{-1}$	$v = kT/h$	$v = 10^{13} \text{ s}^{-1}$	$v = kT/h$	$v = 10^{13} \text{ s}^{-1}$
Al-12Si	59.9	66.7	63.5	70.1	70.0	76.7
Al-33Cu	101.3	107.4	137.7	143.6	131.5	137.4
Al-33Cu-0.4Zr	117.9	124.7	108.8	115.0	116.0	122.2

* For simple cases $v = kT/h$ holds, for others $v = 10^{13} \text{ s}^{-1}$. One or the other value can be obtained based on different assumptions in statistical mechanics.



a Al-12Si; b and d Al-33Cu; c Al-33Cu-0.4Zr

2 Plots of isothermal stress function $f_1(\sigma')$ of given alloys as function of applied stress

sliding in the three aluminium alloys investigated have a common structure.

Figure 5 shows the $g_2(x)$ versus x relationship for the three aluminium alloys. As suggested earlier based on the mathematical form of equation (10), the relationship is indeed nearly linear. The scatter of the experimental points about the best fit curve based on equation (10) is also indicated for different values of x . It can be seen that the fit is reasonable. An average (best fit) value for λ of 5.6021 (or 0.2014 for α) was obtained.

Thus there is reason to believe that the basic units of microscopic boundary sliding in the three aluminium alloys studied had a common structure.

Conclusions

1. Steady state isostructural isothermal superplasticity in the optimal range can be predicted using four experimentally determined, physically meaningful constants described in the text: C_1 , α , σ_m , and σ_0 . From a set of C_1 values at different temperatures, the true activation energy for the rate controlling process ΔU_0 can be determined. The agreement between the experimentally observed strain

rates and those calculated using the above constants is excellent.

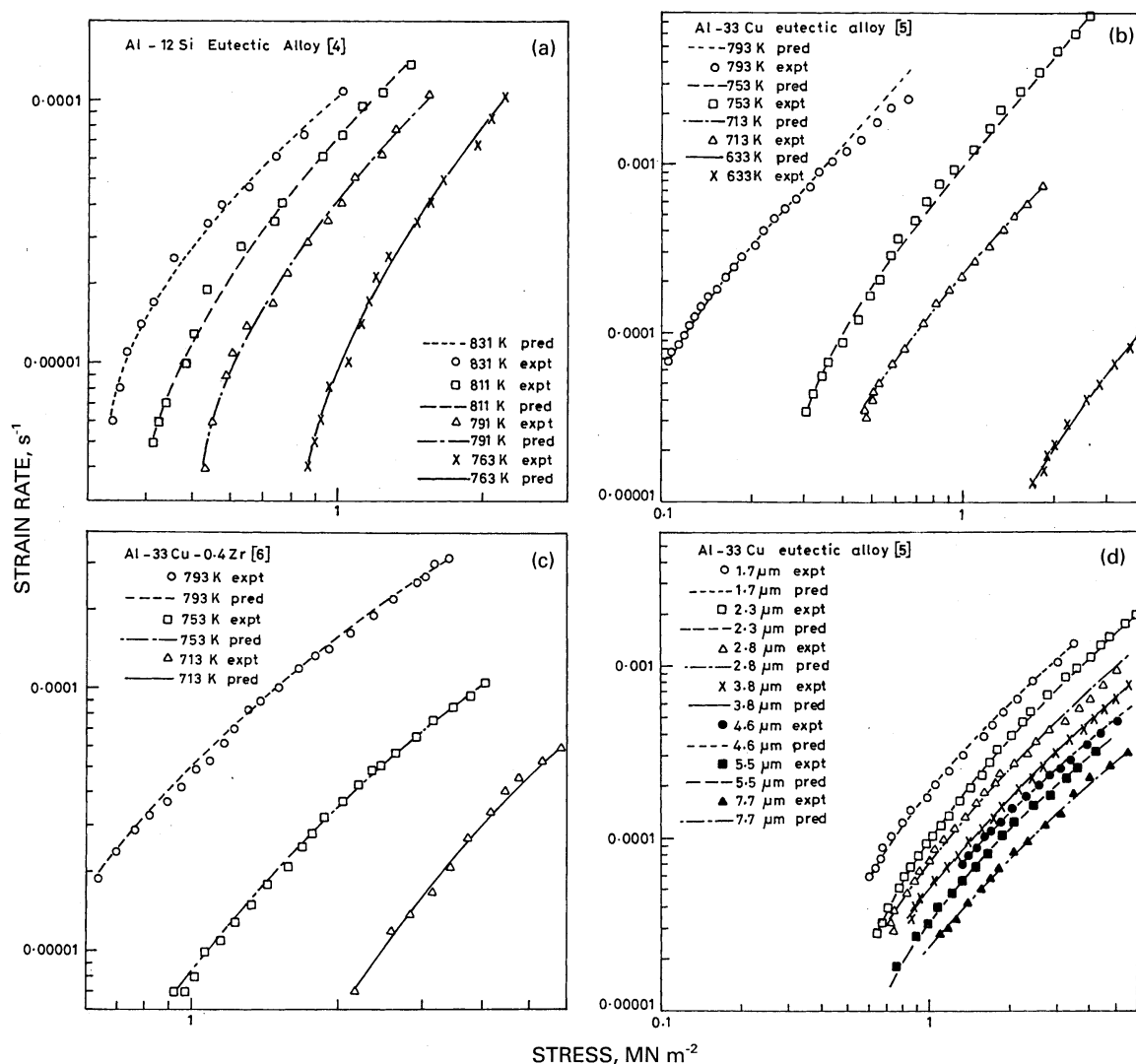
2. From among the 'solutions' obtained from the rate equation by a trial and error procedure, a solution could be found (using a rigorous statistical procedure) for which C_1 varied as $1/L$ and σ_0 as $1/L^a$ (where L is the grain size and a is a constant) as required by the theory presented in Part 1. At the same time excellent estimates of the strain rate of deformation resulted.

3. It was possible to estimate theoretically the values of C_1 , ΔF_0 (ΔU_0), and σ_0 (in full for interphase interfaces and partially for intercrystalline boundaries).

4. On an empirical basis $\sigma_0 = a_0 + a_1/T$, where a_0 and a_1 are constants and T is the absolute temperature of deformation. This dependence is yet to be included in the theoretical framework.

5. The values of ΔU_0 for the rate controlling boundary sliding process for the three alloys analysed here are very similar to those obtained in an earlier phenomenological model and to those obtained by setting $\sigma_0 = 0$ in the present analysis.

6. The internal stress distribution arising from sliding and the stress function that is proportional to the external strain rate have been calculated as functions of the applied stress for the three alloys.



a Al-12Si; b and d Al-33Cu; c Al-33Cu-0.4Zr

3 Comparison between predicted and experimentally measured values of strain rate as function of applied stress corresponding to different experimental conditions for given alloys

7. The apparent viscosity–applied stress relationship has been calculated for the three alloys. The dependence observed of the isothermal apparent viscosity on the stress level and the grain size is understandable because superplastic flow is non-Newtonian viscous ($m \neq 1$).

8. In the three aluminium alloys examined, the basic units of boundary sliding appear to have a common structure.

Appendix

Calculation of ΔF_0 (ΔU_0), C_1 , and σ_0 using model presented in Part 1

FREE ENERGY OF ACTIVATION FOR GRAIN BOUNDARY SLIDING

Elastic energy of 'stress free' shear γ_0 confined to volume V_0 (see equations (2) to (4) of Part 1 and Eshelby⁹) is given by

$$\Delta F_1 = \frac{1}{2} \beta_1 \mu V_0 \gamma_0^2 \quad (12)$$

For an oblate spheroid with $W/2R_0 = 0.5$

$$V_0 = \frac{2\pi}{3} R_0^3$$

and

$$\beta_1 = \frac{1}{3(1-p)} \left[\frac{2\pi}{3^{1/2}} - \frac{5}{2} - \left(\frac{2\pi}{3^{3/2}} - \frac{1}{2} \right) p \right] \quad (13)$$

(See Part 1 for the definition of symbols.)

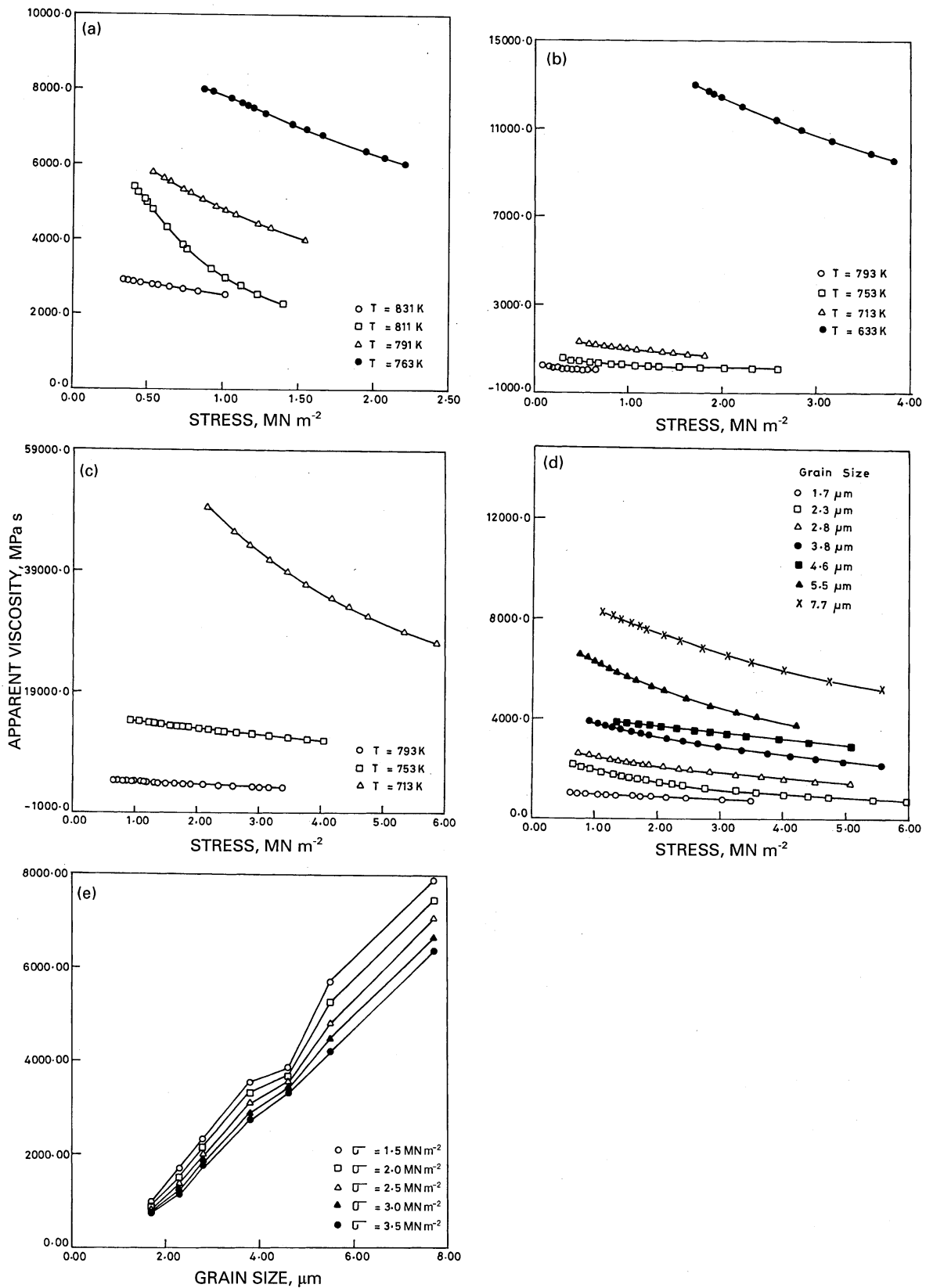
Elastic energy of uniform expansion ε_0 confined to volume V_0 is given by

$$\Delta F_2 = \frac{1}{2} \beta_2 \mu V_0 \varepsilon_0^2$$

where

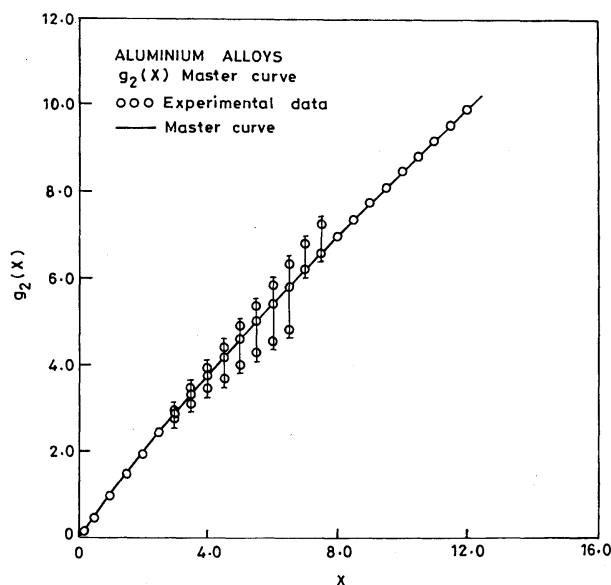
$$\beta_2 = \frac{4}{9} \left(\frac{1+p}{1-p} \right) \quad (\text{independent of shape}) \quad (14)$$

For $p = 1/3$, $\beta_1 = 0.446$ and $\beta_2 = 0.889$. Assuming $W = R_0 = (5/2)b$; $\gamma_0 = 0.1$; $\varepsilon_0 = 0.05$; and $\mu b^3 = 3.5$ eV (6.04×10^{-19} J); $\Delta F_0 (= \Delta F_1 + \Delta F_2) = 0.38$ eV/atom = 39.5 kJ mol⁻¹. Since γ_0 and ε_0 enter the expressions quadratically, any number for ΔF_0 between 10 and 100 kJ mol⁻¹ can be obtained with relatively small deviations from the values actually chosen in the calculations shown above (10% for γ_0 and 5% for ε_0 respectively). These values have been chosen according to existing estimates for metallic glasses and bubble raft experiments.¹⁰ Thus, ΔF_0 has to be determined experimentally. The local strains to be postulated can then be made compatible with the value determined experimentally.



a Al-12Si; b, d, and e Al-33Cu; c Al-33Cu-0.4Zr

4 Calculated apparent viscosity-applied stress relationship for different materials and conditions



5 Curve of $g_2(x)$ versus x for all three aluminium alloys studied (see equation (10))

As $\Delta U_0 = \Delta F_0 + T\Delta S$, it is evident that the values of ΔU_0 reported in Table 4 from experimental results agree very well with the range predicted here theoretically. (As ΔS is absorbed into the fitting parameters, it can not be estimated experimentally.) The system dependence of ΔF_0 in the theoretical calculations comes from μ and γ_0 and the dependence on the shape of the base units is reflected by V_0 and β_1 .

CALCULATION OF C_1

From Part 1

$$C_1 = 2 \frac{W}{L} \frac{\epsilon_0^2 V_0}{kT} \nu \exp(-\Delta F_0/kT) \quad (15)$$

For the present calculations an interatomic spacing b of 3 Å (0.3 nm) in the boundary region is assumed. W is taken as $2.5b$, $V_0 = (4\pi/3)R_0^3(W/2)$ (with $R_0 = W$), and $\epsilon_0 = 0.05$. The ΔF_0 range for estimating the range of C_1 values permissible is taken as 10–100 kJ mol⁻¹.

Based on the experimental data, the range of T values is taken to lie between 650 and 850 K and the grain size range as 1×10^{-6} – 8×10^{-6} m. From the above values and equation (15), C_1 is estimated to lie in the range 2.7×10^{-9} – $0.6959 \text{ Pa}^{-1} \text{ s}^{-1}$. For a ΔF_0 value of 39.5 kJ mol⁻¹, calculated based on results obtained for metallic glasses and bubble raft experiments¹⁰ (see above), C_1 is found to lie in the range 1.94×10^{-4} – $3.74 \times 10^{-4} \text{ Pa}^{-1} \text{ s}^{-1}$.

CALCULATION OF LONG RANGE THRESHOLD STRESS σ_0

From Part 1 and equation (3) of Part 4

$$\sigma_0 = 3^{1/2} (2\mu\gamma_B)^{1/2} \left[\frac{\Delta A/A}{\alpha_f (NA)^{1/2}} \right]^{1/2} \quad (16)$$

The value of μ is calculated from $\mu b^3 = 3.5 \text{ eV}$ ($= 6.04 \times 10^{-19} \text{ J}$) (see above), i.e. $\mu = 2.2 \times 10^{10} \text{ N m}^{-2}$. The specific grain boundary energy γ_B for aluminium is taken as 0.400 J m⁻² (Ref. 11) and $\alpha_1 \approx 1$.

The grains are assumed to have a grain size L and a rhombic dodecahedron shape (often suggested to be close to the shape of real crystals). Then, from the geometry of the rhombic dodecahedron, the ground area per grain on the (111) mesoscopic glide plane $A = (3^{1/2}/4)L^2$.

For the data analysed in this paper L varied between 1×10^{-6} and 8×10^{-6} m. In the calculations N , the number of grain boundaries participating in the mesoscopic sliding event, was allowed to vary between 2 and 10. (Experimental data in this regard are rare; the only result to date (presented in Part 2) gives $N = 4$.)

CASE I: σ_0 AT INTERPHASE BOUNDARY

Based on the geometry of a rhombic dodecahedron and the physical picture for local boundary migration presented in Part 1, the area reduction in the mesoscopic glide plane ΔA is given by

$$\Delta A = \frac{3}{2} qr$$

where q is the projection of the edge of the triangular pyramid of the rhombic dodecahedron on to the mesoscopic (111) glide plane ($= L/3^{1/2}$) and r is the residual misfit left after the completion of non-rate controlling dislocation motion. For the present calculations r has been taken to vary between 0.015 and 0.15 nm.

CASE II: σ_0 AT INTERCRYSTALLINE BOUNDARY

Here local boundary migration takes place by diffusion. Then

$$\Delta A = \frac{3}{2} q(h - \delta)$$

where h is the height of the pyramid to be flattened by diffusion (for a rhombic dodecahedron equal to $(6^{1/2}/12)L$) and δ is the grain boundary width (taken, as before, as equal to $2.5b$ where $b = 0.3 \text{ nm}$).

Therefore

$$\Delta A = \frac{2^{1/2}}{8} L^2 [1 - 2\delta(6^{1/2}/L)]$$

Results

For case I, a σ_0 range of 0.34–1.26 MN m⁻² is predicted. This is generally in good agreement with the experimental values obtained by numerical solution (Tables 1 and 3) that lie predominantly in the range 0.25–1.2 MN m⁻².

For case II, a σ_0 range of 36.0–149.4 MN m⁻² is predicted. This is an interesting result that reveals the following.

1. Resistance to plane interface formation by local boundary migration at the intercrystalline boundaries of a material of uniform grain size and shape is extremely high. This is because in this case ΔA , the area to be removed by diffusion, is equal to 39.3–40.8% of the total ground area of a grain in the mesoscopic glide plane. This explains why the grain coarsening resistance of materials is high when the grain size distribution is very narrow and the shapes of the grains are similar.

2. When the magnitude of the external stress plus the internal stress resulting from boundary sliding is large enough to force lattice dislocations into the grains from grain boundary bends (see Part 1), this mechanism of local boundary migration will operate more readily than the mechanism involving diffusion even at an intercrystalline boundary.

3. In superplastic alloys made up of grains of non-uniform sizes and shapes, the long range threshold stress needed for mesoscopic sliding is considerably less than that predicted by equation (16) (see, e.g. Tables 1 and 3).

For an intercrystalline boundary, when the grain size is

uniform

$$\frac{\Delta A/A}{\alpha_f(NA)^{1/2}}$$

varies as $L^{-1/2}$ (see above).

But in Table 3 it is seen that experimentally σ_0 varies as $L^{-0.12}$. Using this result, if equation (16) is modified as

$$\sigma_0 = 3^{1/2}(2\mu\gamma_B)^{1/2} \left[\frac{\Delta A/A}{\alpha_f(NA)^{1/2}} \right]^a; \quad a = 0.12 \quad \dots (17)$$

a σ_0 range of 0.77–1.09 MN m $^{-2}$ is predicted, which is in accordance with the experimental values obtained by a numerical best fit solution.

At this stage, it has not been possible to show that $a \approx 0.12$ for real materials of non-uniform grain shapes and sizes or for the situation where intercrystalline as well as interphase boundaries align together to form a plane interface. However, a parallel between this result and the empirical modification made to the classical equation predicting grain growth can be drawn. There also, the predicted grain size increase is as $L^{-1/2}$. This is then empirically modified to L^{-a} , with $a \approx 0.18, 0.25$, etc. according to the system considered.

Thus, at the present stage of development, the theory is unable to reveal *a priori* the values of σ_m , α , and a . However, σ_m and α may be obtained by examining the basic unit of microscopic sliding using molecular dynamics and a may be obtained by analysing the problem of local boundary migration in real materials of non-uniform grain sizes and shapes.

This situation appears reasonably practicable when it is noted that the alternative Dorn equation used to explain steady state isostructural superplasticity in two phase alloys requires 10 adjustables, most of which can not be calculated from theory. In addition, regions I and II are attributed to two different mechanisms.

Acknowledgements

This paper is dedicated to Professor C. V. Sundaram for having helped nucleate superplasticity research at IIT, Madras.

The authors thank Professors R. Subramanian and P. R. Parthasarathy for a useful discussion on the statistical techniques adopted in this paper. Financial support for the work was derived from the Indo-US Project No. DST/INT/USHRF/211/88 of the Department of Science and Technology, Government of India Grant No. N00014-93-1-0041/R&T Project Code: US-India 0501 of the Office of Naval Research, Arlington, VA, USA. One of the authors (KAP) thanks the Alexander-von-Humboldt-Stiftung, Bonn, for financial support.

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CORRIGENDA

The following corrections should be made to Part 1 in this series of papers 'Model for grain boundary sliding and its relevance to optimal structural superplasticity', *MST*, May 1996, Volume 12, pp. 391–399

Page 393

In the fourth line of the right hand column

τ_i should replace τ_1

In equation (13)

τ_i should replace τ_0

In the line immediately below equation (13)

$\tau \leq \tau_i$ should replace $\tau \leq \tau_1$

Page 394

The last sentence in the left hand column should read:

'No specific shape of the grains is assumed because expressions for mesoscopic boundary sliding (and hence structural superplasticity) can be derived independent of the grain shape.'

In the ninth line of the section on 'Optimal superplastic deformation'

$g(\sigma_i)$ should replace $g(\sigma_1)$